La Scuola d'Italia / Dr. Huson / IB Math　　　　　　Name:
15 April 2026

5.13 Classwork: Exponential Differentiation (no calculator)

1. [Maximum mark: 4]

   The function g is defined by $g(x) = e^{x^2+1}$, where $x \in \mathbb{R}$.

   Find $g'(-1)$.

2. [Maximum mark: 16]

   Let $y = \dfrac{\ln x}{x^4}$ for $x > 0$.

   (a) Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1 - 4\ln x}{x^5}$. [3]

   Consider the function defined by $f(x) = \dfrac{\ln x}{x^4}$ for $x > 0$ and its graph $y = f(x)$.

   (b) The graph of f has a horizontal tangent at point P. Find the coordinates of P. [5]

   (c) Given that $f''(x) = \dfrac{20\ln x - 9}{x^6}$, show that P is a local maximum point. [3]

   (d) Solve $f(x) > 0$ for $x > 0$. [2]

   (e) Sketch the graph of f, showing clearly the value of the x-intercept and the approximate position of point P. [3]